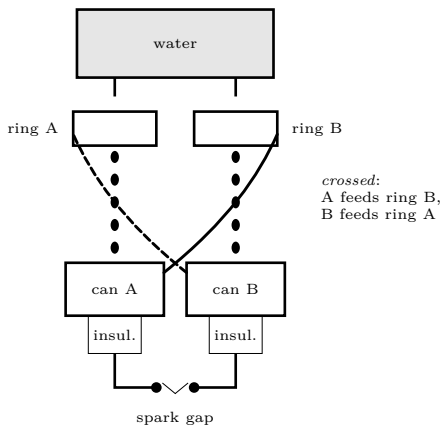


Demo: Kelvin Water Dropper

Make sparks out of falling water and nothing else

Lesson Demo B · 45 minutes to build · after Lesson 1 — no external power at all

What you need	
- Two metal cans, tops and bottoms removed (the rings) - Two more cans as collectors, or two plastic cups lined with foil - A plastic bottle or bucket with two small holes for drip-	- pers - Insulating supports: plastic cups, dry wood, foam - Bare wire for the cross-connections - Water, ideally slightly salty



Predict first	
Water is not charged. There is no battery anywhere in this apparatus. So where could the energy for a spark possibly come from? Commit to an answer, then look at the height the water falls from.	
The seed	One side starts with a tiny random charge imbalance — there is always some.
Induction does the rest	Say can A is slightly negative. It is wired to ring B, so ring B is slightly negative, which repels electrons from the drops forming inside it. Those drops fall slightly positive into can B. Can B is wired to ring A, making it positive, so drops through ring A fall negative into can A — making A more negative than it started.
Runaway	Each side makes the other side stronger. The charge grows exponentially until the air between the cans breaks down. This is positive feedback made visible.
Where the energy comes from	Gravity. Each drop is pulled away from a can that is attracting it, so falling water does work against the electric field. It is a generator whose fuel is height.
The voltage	Thousands of volts, easily — a visible, audible spark. The current is microscopic, so the shock is a sharp sting at worst.

Making it actually work

Insulate everything	This is the whole game. The cans must sit on genuinely dry insulators, and no wire may touch the frame. Charge leaks away faster than it is generated otherwise, and nothing happens.
Dry day	Humid air conducts. On a muggy August afternoon this will not work at all — which is itself the lesson from Lesson 1.
Cross the wires	Can A to ring B, can B to ring A. Wire them straight across instead and it will never start. Get somebody to check this before you troubleshoot anything else.
Drops, not a stream	The stream must break into separate drops <i>inside</i> the ring. A continuous jet conducts the charge straight back and defeats it.
Be patient	It can take thirty seconds to a couple of minutes to build to the first spark. Then it fires repeatedly.

Why this is the best demonstration in the course

It uses no electricity to make electricity. A jug of water on a shelf produces visible sparks, and the only inputs are gravity and one accidental electron. Lord Kelvin built this in 1867. It teaches induction, insulation, positive feedback and energy conservation in one object — and it is genuinely startling the first time it cracks.

This lesson is low voltage. Everything here runs from batteries or a small plug-in supply and cannot hurt you. That changes at Lesson 10. Build the habits now — one hand, discharge before touching, check before you power up — while the stakes are nothing, so that they are automatic when the stakes are real.

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